

A Compact Bias-Free ConvNeXt U-Net for Blind Gaussian Image Denoising

Small, strictly bias-free, and blind: how far a compact ConvNeXt U-Net reaches on the color-denoising leaderboard

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Abstract

We describe and evaluate BF-CONVUNEXT, a compact bias-free CONVNEXT U-Net for blind additive-white-Gaussian-noise (AWGN) image denoising. The network combines four ingredients, none individually novel. A frozen depthwise Gabor stem gives an oriented band-pass front end with zero trainable parameters. A Laplacian-pyramid encoder routes the full-resolution high-frequency residual into the skip connection, with three extra CONVNEXT blocks per skip band. A CONVNEXT-V1 U-Net forms the body. The construction is bias-free from end to end: every convolution has no additive bias, the head is linear, activations are LeakyReLU, and normalization is a variance-only batch norm. Together these make the whole 3.39M-parameter network exactly degree-1 homogeneous at inference, $D(\alpha y) = \alpha D(y)$. That homogeneity is what licenses the Miyasawa/Tweedie reading of the denoiser residual as a scaled score — a *local* score estimate, as our non-conservativity result later makes precise, not a global prior — and, following Mohan et al., supports blind generalization across noise levels from a single model. We train one blind model with a linear noise- σ curriculum spanning $\sigma_{255} \approx 6$ to 127 and evaluate it, unchanged, against the published color-denoising leaderboard on all four standard sets (CBSD68, Kodak24, McMaster, Urban100) at $\sigma \in \{15, 25, 50\}$ under the full-image protocol. The single blind model beats DnCNN and FFDNet on every set at every noise level (on average +0.7 dB over DnCNN; FFDNet is non-blind yet still trails), and sits a small, consistent margin behind the heavyweight CNN/transformer state of the art: about 0.4 dB on CBSD68 and Kodak24, 0.7 dB on McMaster, and 1.0 dB on Urban100, at roughly one tenth of their parameters. We state the limits once and move on: the homogeneity is inference-only and checkpoint-specific, and the learned residual is a *local* score, not a global prior (its Jacobian is non-conservative), so plug-and-play/RED and calibrated-posterior guarantees do not transfer. That same local score is nonetheless enough to drive stochastic sampling from the implicit prior and to solve linear inverse problems — inpainting, super-resolution, deblurring, compressive sensing. Results are a single checkpoint, reported in PSNR (plus SSIM for our own model), without ablations.

1 Introduction

Blind Gaussian image denoising recovers a clean image x from $y = x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, with σ unknown at test time. It is both a canonical low-level vision task and a common building block inside modern inverse-problem solvers. Two facts frame this paper. First, the field’s accuracy is dominated by large transformer/CNN hybrids (DRUNet, SwinIR, Restormer, SCUNet) with tens of millions of parameters, often trained per-noise-level. Second, a much older and simpler observation remains underexploited in a modern architecture: removing all additive constants from a denoising CNN makes it exactly scale-equivariant, and therefore able to generalize blindly across noise levels that a biased network cannot handle [Mohan et al., 2020].

This paper takes the second fact seriously and asks one empirical question: how far does a compact, strictly bias-free, modern-conv-block U-Net get on the standard color-denoising leaderboard as a single blind model? Our answer is BF-CONVUNEXT, a 3.39M-parameter bias-free CONVNEXT U-Net that composes four ideas we did not invent:

- a frozen depthwise Gabor stem [Gabor, 1946]: a bank of oriented, frequency-tuned band-pass filters fixed at initialization (zero trainable parameters), followed by a single learned 1×1 mixing projection;

- a Laplacian-pyramid encoder [Burt & Adelson, 1983] that, at each downsampling junction, sends the full-resolution high-frequency residual band to the skip connection and passes the low-frequency band downward, with three extra CONVNEXT blocks applied to each high-frequency skip band;
- a CONVNEXT-V1 U-Net body [Liu et al., 2022, Ronneberger et al., 2015] (depthwise 7×7 conv, inverted-bottleneck MLP, LayerScale);
- bias-freedom from end to end: no additive bias in any layer, a linear head, LeakyReLU activations, and a variance-only batch normalization, chosen together so the network is globally degree-1 homogeneous at inference.

None of these pieces is new. The contribution is the specific, carefully bias-preserving way they are combined so that homogeneity survives end to end, together with an apples-to-apples measurement of the result against the published leaderboard on the same test sets and full-image protocol. The result is favorable: the compact blind model uniformly beats the classic sub-1M CNNs and lands about half a dB behind models an order of magnitude larger.

1.1 What this paper is (and is not)

This is an integration-and-evaluation paper, closer to an engineering report than to a novel-component paper. It makes three contributions:

1. a precise description of BF-CONVUNEXT, covering the exact topology, the frozen Gabor stem, the Laplacian-pyramid skip decomposition, the CONVNEXT-V1 block, and the bias-free construction that makes the whole network degree-1 homogeneous, at a level of detail sufficient to reimplement it;
2. a fair benchmark: one blind checkpoint, evaluated unchanged on all four standard color sets at three noise levels under the full-image protocol, tabulated against six published baselines;
3. an explicit account of the theory’s limits: the Miyasawa/Tweedie residual-as-score reading is only a motivation, and we show (Section 6) that the learned operator is non-conservative, so it is not the gradient of any global prior.

Correspondingly, it is not an ablation study — we do not isolate each component’s contribution here (Section 6) — and the leaderboard comparison is in PSNR, since the collated source is PSNR-only (we add SSIM for our own model, Table 2). The theory limits of contribution (iii) are developed once, in Section 3.

2 Architecture

BF-CONVUNEXT is a two-level U-Net operating on images normalized to $[-0.5, +0.5]$. All parameter counts and layer shapes below are read from the actual trained checkpoint (`convunext_denoiser_base`, 3,393,936 parameters total: 3,378,540 trainable and 15,396 frozen). Figure 1 shows the data flow.

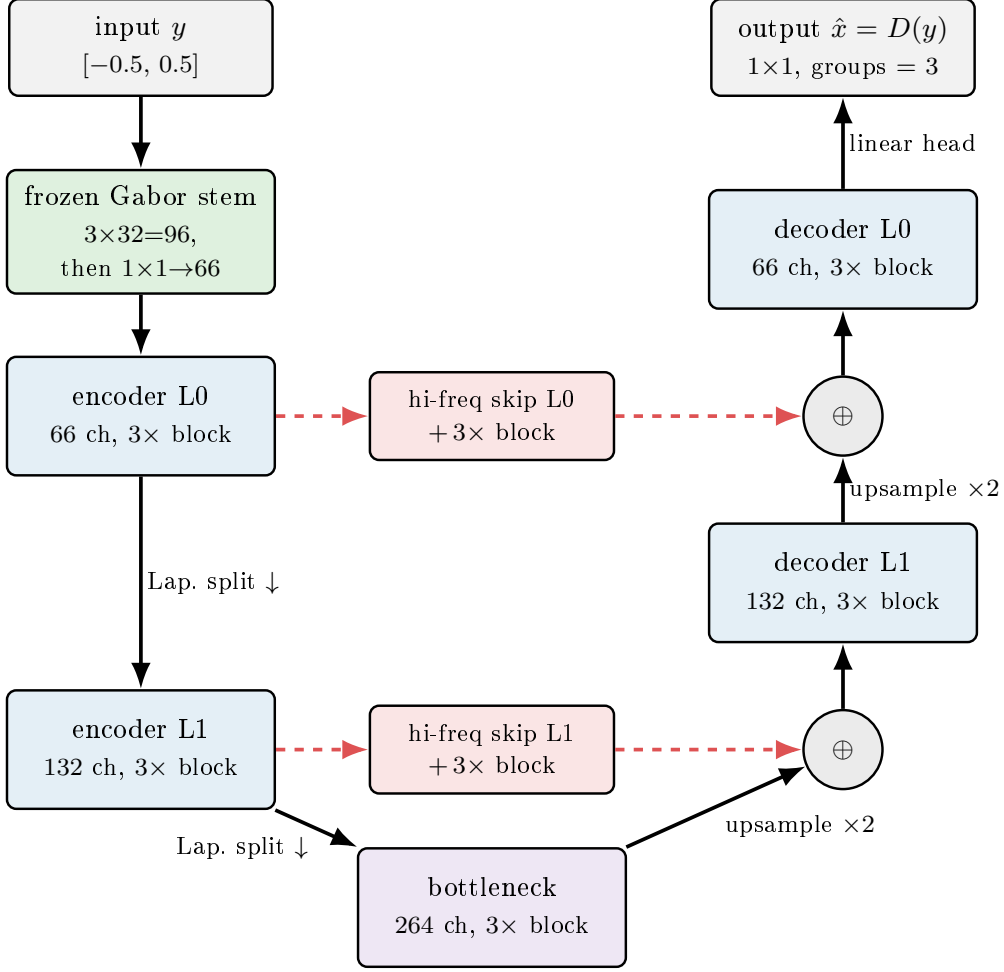


Figure 1: BF-CONVUNEXT data flow (channel widths shown). The encoder (left) descends, the decoder spine (right) ascends, and the bottleneck sits at the base. A frozen Gabor stem feeds a two-level CONVNEXT-V1 U-Net. Downsampling is a Laplacian-pyramid split: the low-frequency band descends (solid arrows) while the full-resolution high-frequency residual becomes the skip (dashed), passing through three extra bias-free CONVNEXT blocks. At each decoder level the skip and the upsampled feature are first combined at an explicit parameter-free, channel-matched addition ($\oplus$), a distinct step, and only then processed by the decoder block. The head is a bias-free grouped 1×1 convolution (3 groups). Every operation is degree-1 homogeneous at inference, so $D(\alpha y) = \alpha D(y)$.

2.1 U-Net topology

The channel widths follow $c_i = \text{round}(66 \cdot 2^i)$ for level i , giving $66 \rightarrow 132$ in the encoder, 264 at the bottleneck, and $132 \rightarrow 66$ in the decoder. Each of the five scopes (two encoder levels, bottleneck, two decoder levels) contains three CONVNEXT-V1 blocks. Upsampling is bilinear ($\times 2$). Channel counts are matched by a parameter-free `MatchChannels` operator (zero-pad to widen, slice to narrow) rather than a learned 1×1 projection, and decoder skips are fused by addition, $\text{dec} \leftarrow \text{Add}(\text{skip}, \text{MatchChannels}(\text{up}))$, not by concatenation. Both choices keep the merge exactly linear and homogeneous.

2.2 Frozen Gabor stem

In a plain CNN the first learnable operation on a noisy image is a small learned convolution. Here it is replaced by a fixed oriented band-pass front end. A depthwise convolution with 32 filters per input channel and a 7×7 kernel is initialized with a Gabor bank, each output channel being one 2D Gabor wavelet

$g(u, v) = \exp\left(-\frac{u_\theta^2 + \gamma^2 v_\theta^2}{2s^2}\right) \cos\left(\frac{2\pi u_\theta}{\lambda} + \psi\right)$ at a swept orientation, frequency, and phase, and is marked non-trainable, so it contributes no trainable parameters and never adapts. With three input channels it emits $3 \times 32 = 96$ feature maps, which a single learned bias-free 1×1 convolution mixes down to the 66 stem channels. The rationale is standard signal processing: an oriented multi-scale band-pass decomposition is a strong, noise-agnostic image representation that the network need not spend capacity (or risk overfitting noise) to relearn.

2.3 Laplacian-pyramid skips and high-frequency blocks

Instead of max-pooling, each encoder junction performs a Laplacian-pyramid split. Given the level input z , a fixed Gaussian blur (5×5) and stride-2 blur-pool produce the low band $\ell = \text{blurpool}(\text{gauss}(z))$, and the high band is the full-resolution residual $h = z - \text{up}(\ell)$. The split is lossless ($\text{up}(\ell) + h = z$) and channel-preserving. The low band ℓ descends to the next level. The high band h becomes the skip connection, but only after passing through three additional bias-free CONVNEXT-V1 blocks (`skip_highfreq`), applied at both encoder levels. This is a wavelet-shrinkage-style inductive bias. Gaussian noise is spectrally flat while natural-image structure concentrates at low frequencies, so forcing the decoder to reconstruct from an explicit low-band descent plus a processed high-band residual denies it the trivial full-resolution identity-copy shortcut and matches the signal/noise spectral structure of the task.

2.4 The CONVNEXT-V1 block

Each block is a residual branch with call order

$$x \mapsto x + \gamma \odot \left[W_{\downarrow} \phi \left(W_{\uparrow} \text{N}(\text{DW}_{7 \times 7} x) \right) \right],$$

where $\text{DW}_{7 \times 7}$ is a depthwise 7×7 convolution, N is the normalization (Section 2.5), $W_{\uparrow}$ is a 1×1 expansion to $4 \times$ width, ϕ is LeakyReLU with negative slope 0.1 (followed by dropout 0.1 at training time), $W_{\downarrow}$ is a 1×1 reduction, and γ is a per-channel LayerScale multiplier (initialized 10^{-4} , floored at 10^{-6}). All convolutions are bias-free. We use the V1 block rather than V2, because the V2 variant inserts a Global Response Normalization whose learnable additive β is bias-like and would break strict bias-freedom.

2.5 Bias-free construction and degree-1 homogeneity

The defining property of BF-CONVNEXT is that the entire map D is positively homogeneous of degree one at inference:

$$D(\alpha y) = \alpha D(y) \quad \text{for all } \alpha > 0. \tag{1}$$

This holds because every constituent operation is linear-through-the-origin or degree-1 homogeneous, and no operation injects an additive constant:

- every convolution (Gabor stem, 1×1 mix, all block convolutions, the head) has `use_bias=False`;
- the head activation is linear ($f(x) = x$);
- LeakyReLU is positively homogeneous, $\text{LReLU}(\alpha x) = \alpha \text{LReLU}(x)$ for $\alpha > 0$ (this is why LeakyReLU is used rather than GELU, which is not homogeneous);
- normalization is a *variance-only* batch norm, `BIASFREEBATCHNORM`: it has no running mean and no β offset, only a fixed non-trainable running variance, so at inference $\text{N}(x) = \gamma x / \sqrt{\text{Var} + \epsilon}$ is exactly linear in x ;
- the Laplacian split, `MatchChannels`, bilinear upsampling, and additive skip merge are all linear.

The choice of `BIASFREEBATCHNORM` over the CONVNEXT default LayerNorm matters and is not cosmetic. LayerNorm divides by a per-sample statistic that itself scales with the input, making it scale-invariant (degree-0), which would destroy Eq. (1) for the whole network. One scoping caveat is worth flagging, since most treatments omit it: homogeneity is an inference-time property. During training `BIASFREEBATCHNORM` uses the per-batch variance, which makes it degree-0, and Eq. (1) holds only once the running statistics are frozen.

3 Why bias-free: Miyasawa/Tweedie, and its limits

Bias-freedom is more than an aesthetic choice: it connects the trained denoiser to a score, and it lets one blind model span a wide noise range. We state the motivation first, then its boundaries.

Miyasawa/Tweedie. For $y = x + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ and marginal density $p_\sigma(y) = \int \mathcal{N}(y; x, \sigma^2 I) p(x) dx$, the minimum-MSE estimator obeys the Miyasawa/Tweedie identity

$$\hat{x}(y) = \mathbb{E}[x | y] = y + \sigma^2 \nabla_y \log p_\sigma(y). \quad (2)$$

Hence for an MMSE-optimal D the denoiser residual $f(y) = D(y) - y$ is exactly σ^2 times the score $\nabla_y \log p_\sigma(y)$ of the noise-blurred density. A plain mean-squared-error objective is therefore well motivated: it drives D toward the MMSE estimator whose residual is a scaled score. This is why BF-CONVUNEXT is trained with MSE and read as an implicit-prior score model — with the caveat, made precise in Section 3, that the *learned* residual realizes this only as a *local* score estimate, not the gradient of a global density.

Homogeneity buys blind generalization. Mohan et al. [2020] observe that a bias-free denoiser is degree-1 homogeneous (Eq. (1)) and therefore behaves as a locally linear, input-adaptive filter $D(y) \approx J(y) y$ whose Jacobian $J(y) = \partial D / \partial y$ is degree-0 (scale-invariant). Because J depends only on the direction of y , not its magnitude, a network trained over one range of noise levels extrapolates gracefully to unseen levels. A biased network, whose additive terms fix an absolute scale, instead collapses off its training range. This is the property that makes a single blind model viable, and it is the empirical target of our noise curriculum (Section 4).

What homogeneity does not buy. It is easy to over-read Eq. (2), so we state two boundaries explicitly.

1. **Homogeneity is checkpoint-specific, not an architectural guarantee.** A network is exactly homogeneous only if every operation is genuinely bias-free. For this checkpoint the relative homogeneity error $\|D(\alpha y) - \alpha D(y)\| / \|\alpha D(y)\|$ is at the level of floating-point round-off across $\alpha \in \{\frac{1}{4}, \frac{1}{2}, 2, 4, 8\}$, yet sibling checkpoints of the same family have shown deviations up to about 0.14 when a single non-homogeneous operation slips in. Eq. (1) must be verified per checkpoint, not assumed.
2. **The learned residual is not a conservative field.** Eq. (2) would make $f(y)$ the gradient of the scalar $\sigma^2 \log p_\sigma(y)$ only if the Jacobian $J(y)$ were symmetric, and it is not. On trained checkpoints of this family the measured Jacobian asymmetry $\|J - J^\top\| / \|J\|$ is $\mathcal{O}(0.5)$, roughly four orders of magnitude above a symmetric-blur control, so no global energy or log-density function exists whose gradient is the residual. Plug-and-play and Regularization-by-Denoising convergence guarantees assume a conservative, symmetric-Jacobian denoiser, so they do not transfer here, and the residual should be read as a useful local score estimate rather than a calibrated global prior.

We return to the practical consequences in Section 6. (This non-conservativity is a statement about a *global* energy; it does not prevent well-calibrated *marginal* per-pixel intervals, which we obtain by conformal prediction in Section 5.6.)

4 Experimental protocol

Data and noise curriculum. We train on 256×256 patches drawn from COCO (train2017) and DIV2K (train), sourced with a per-directory weighting so the $\sim 118\text{K}$ COCO images do not drown DIV2K’s ~ 800 . Patches are normalized to $[-0.5, +0.5]$ (peak-to-peak range 1.0) and corrupted with additive Gaussian noise whose per-patch standard deviation is drawn uniformly from $[0, \sigma_{\max}]$, where the upper bound $\sigma_{\max}$ is widened linearly over training by a curriculum callback from 0.025 to 0.5 across 100 epochs. Starting narrow and widening stabilizes early optimization, and bias-freedom then supplies the cross-level generalization.

The σ convention (stated precisely). The model operates entirely in $[-0.5, +0.5]$, so its noise level $\sigma_{\text{norm}} \in [0, 0.5]$. The literature reports noise on the $[0, 255]$ pixel scale, and the two are related by $\sigma_{255} = \sigma_{\text{norm}} \cdot 255$. The curriculum’s $\sigma_{\text{max}} : 0.025 \rightarrow 0.5$ therefore spans $\sigma_{255} \approx 6.4 \rightarrow 127.5$, which covers and exceeds the classic 15/25/50 benchmark regimes as a single blind model. All PSNR values are computed with `max_val = 1.0` (matched to the peak-to-peak range). Because PSNR is scale-invariant under a matched range, these dB numbers are directly comparable to published `max_val = 255` results, with no further conversion.

Optimization. Loss is plain MSE (the Miyasawa-optimal objective, Section 3). We use AdamW (learning rate 10^{-3} , cosine decay, 10 warmup epochs, decoupled weight decay 4×10^{-3}), gradient-norm clipping at 1.0, batch size 4, 100 epochs, full `fp32` (mixed precision was measured slower on this convolution-heavy U-Net because the bilinear-upsample gradient forces XLA off).

Evaluation. We report two evaluations of the single trained checkpoint. (i) A blind PSNR-vs-noise sweep on DIV2K-validation (100 patches at 256px). (ii) The standard leaderboard comparison on all four color sets (CBSD68, Kodak24, McMaster, Urban100) at $\sigma \in \{15, 25, 50\}$ using the full-image protocol (whole images, reflect-padded to a multiple of 16), the same protocol the published numbers use. We additionally report a post-hoc per-pixel split-conformal coverage study. The two evaluations draw different samples (the sweep uses 100 DIV2K patches; the coverage study, 500 held-out COCO crops), so their PSNR columns are independent measurements, not cross-checks of one another.

5 Results

5.1 Blind denoising across noise levels

Figure 2 shows the blind PSNR-vs-noise sweep on DIV2K-validation. A single model, given no noise-level input, denoises across the whole curriculum range, from 40.7 dB at $\sigma_{255}=5$ down to 27.2 dB at $\sigma_{255}=100$ (well beyond the 15/25/50 benchmark regimes), improving the noisy input by +6.4 to +16.7 dB with no cliff at the high-noise end. (Gains are relative to the noisy input *as fed to the model* — clipped to the $[-0.5, +0.5]$ domain, which at high σ has a higher PSNR than the unclipped Gaussian, so the improvement is smaller than $20 \log_{10}(255/\sigma_{255})$ would suggest.)

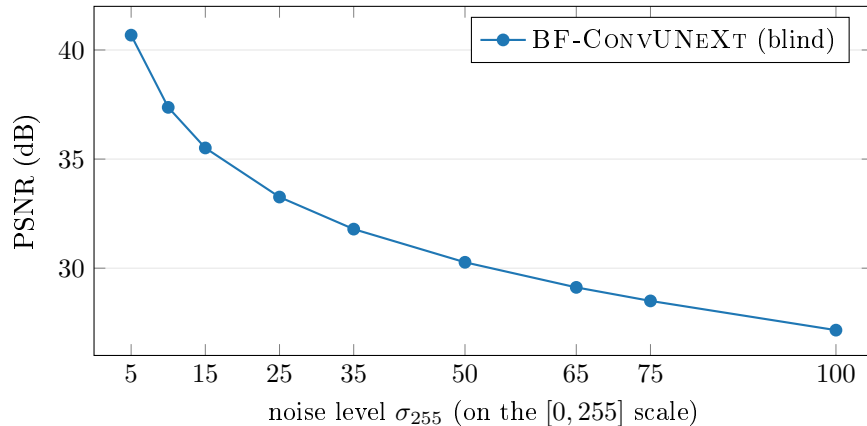


Figure 2: Blind PSNR vs. noise on DIV2K-validation (100 patches, 256px). One model spans $\sigma_{255} \approx 5$ –100 with no noise-level input and degrades gracefully well past the classic 15/25/50 regimes. The 95% bootstrap CIs (not shown) are ± 0.4 – 0.8 dB.

5.2 Comparison against the color-denoising leaderboard

Table 1 places BF-CONVUNEXT against six published baselines on all four standard color sets at $\sigma \in \{15, 25, 50\}$, under the identical full-image protocol and PSNR convention. The pattern is uniform: the single 3.39M blind model beats DnCNN and FFDNet on every set at every noise level, and trails the heavyweight state of the art by a small, consistent margin that widens on Urban100.

Table 1: Color AWGN denoising, PSNR (dB), full-image protocol. BF-CONVUNEXT is a single 3.39M-parameter blind model (no noise-level input). Baseline numbers are the published values collated by Zhang et al. [2022]. **Bold** marks where BF-CONVUNEXT beats a given baseline. Approximate parameter counts are listed in the header; DnCNN/FFDNet are < 1M, the four rightmost are 12–33M.

Set	σ_{255}	BF-CONVUNEXT (3.4M)	DnCNN (0.6M)	FFDNet (0.5M)	DRUNet (32M)	SwinIR (12M)	Restormer (26M)	SCUNet (18M)
CBSD68	15	33.95	33.90	33.87	34.30	34.42	34.40	34.40
CBSD68	25	31.35	31.24	31.21	31.69	31.78	31.79	31.79
CBSD68	50	28.18	27.95	27.96	28.51	28.56	28.60	28.61
Kodak24	15	34.97	34.60	34.63	35.31	35.34	35.47	35.34
Kodak24	25	32.56	32.14	32.13	32.89	32.89	33.04	32.92
Kodak24	50	29.51	28.95	28.98	29.86	29.79	30.01	29.87
McMaster	15	34.79	33.45	34.66	35.40	35.61	35.61	35.60
McMaster	25	32.61	31.52	32.35	33.14	33.20	33.34	33.34
McMaster	50	29.61	28.62	29.18	30.08	30.22	30.30	30.29
Urban100	15	34.20	32.98	33.83	34.81	35.13	35.13	35.18
Urban100	25	31.92	30.81	31.40	32.60	32.90	32.96	33.03
Urban100	50	28.81	27.59	28.05	29.61	29.82	30.02	30.14
<i>avg</i> $\sigma=15$		34.48	33.73	34.25	34.96	35.13	35.15	35.13
<i>avg</i> $\sigma=25$		32.11	31.43	31.77	32.58	32.69	32.78	32.77
<i>avg</i> $\sigma=50$		29.03	28.28	28.54	29.52	29.60	29.73	29.73

Averaged over the four sets, BF-CONVUNEXT is +0.75/+0.68/+0.75 dB above DnCNN at $\sigma = 15/25/50$ and +0.23/+0.34/+0.49 dB above FFDNet (which, unlike our model, is *non-blind* and receives a noise-level map). It is −0.48/−0.47/−0.49 dB relative to DRUNet and −0.67/−0.67/−0.70 dB relative to Restormer, the leaderboard top. The gap is smallest on CBSD68 and Kodak24 (~ 0.4 dB), larger on McMaster (~ 0.7 dB), and largest on Urban100 (~ 1.0 dB). Urban100’s repetitive, self-similar structures are exactly where the transformers’ non-local attention (SwinIR, Restormer) and DRUNet’s large receptive field have the most to exploit and a compact local conv U-Net the least.

5.3 Structural similarity (SSIM)

Table 2 reports SSIM alongside PSNR for BF-CONVUNEXT on the same four sets and full-image protocol (same checkpoint as Table 1), computed on the $[0, 1]$ representation with a data range of 1.0. SSIM tracks PSNR’s ordering across noise, highest at $\sigma_{255}=15$ (average 0.930) and degrading gracefully to average 0.827 at $\sigma_{255}=50$. One cross-metric wrinkle is worth noting: Urban100 has the highest SSIM of the four sets at every noise level (0.944/0.918/0.866) despite its middling PSNR. Its regular, high-contrast man-made structures, the same self-similarity that costs PSNR against non-local models, yield strong local structural agreement once denoised. We report SSIM for BF-CONVUNEXT only, because our comparison source [Zhang et al., 2022] tabulates PSNR alone and SSIM values assembled from mixed sources use inconsistent windowing, border, and color conventions, so a baseline SSIM column would be an unreliable comparison.

Table 2: BF-CONVUNEXT PSNR (dB) and SSIM on the four color benchmark sets, full-image protocol, single blind checkpoint (identical protocol and model as Table 1; the PSNR columns reproduce it exactly). SSIM on $[0, 1]$, data range 1.0. Baseline SSIM is not tabulated (see Section 5.3).

Set	$\sigma_{255} = 15$		$\sigma_{255} = 25$		$\sigma_{255} = 50$	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
CBSD68	33.95	0.9311	31.35	0.8873	28.18	0.8010
Kodak24	34.97	0.9263	32.56	0.8870	29.51	0.8112
McMaster	34.79	0.9181	32.61	0.8879	29.61	0.8287
Urban100	34.20	0.9437	31.92	0.9182	28.81	0.8661
<i>average</i>	34.48	0.9298	32.11	0.8951	29.03	0.8268

5.4 Qualitative examples

Figure 3 shows the model on three held-out benchmark images at $\sigma=25$ — a Kodak24 scene, an Urban100 facade (fine repetitive structure, a known denoiser stress case), and a McMaster crop (saturated colour). Full-image inference is run under the paper’s protocol; the panels are a detail crop of the result. Noise is removed cleanly while edges, repeated structure, and colour transitions are preserved; per-image PSNR/SSIM (computed on the full image) are annotated.

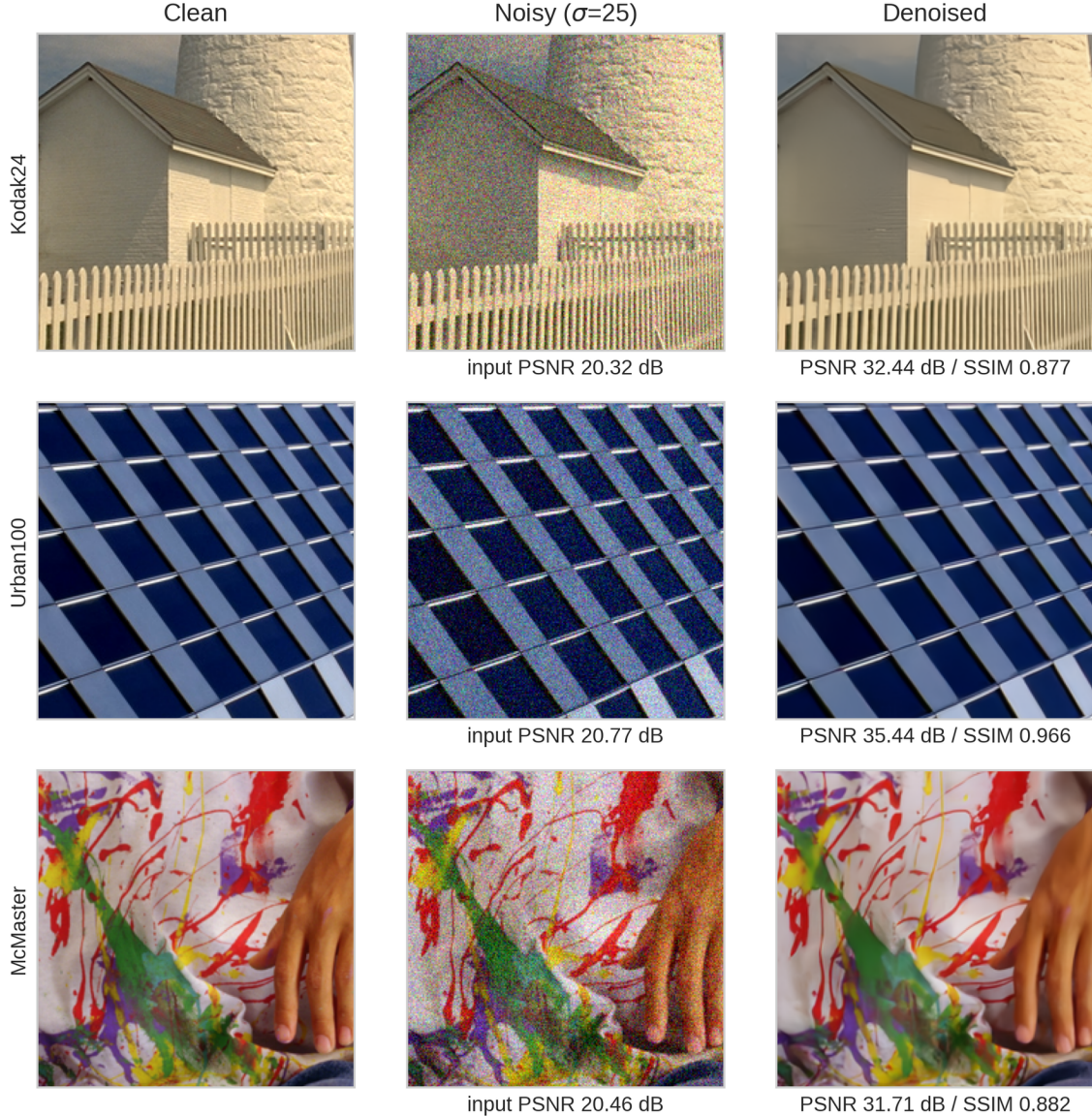


Figure 3: Clean / noisy ($\sigma_{255}=25$) / denoised on three held-out benchmark images (rows: Kodak24, Urban100, McMaster). Per-image PSNR and SSIM (full image, max_val=1) are annotated under each panel; the blind model receives no noise-level input.

5.5 Accuracy for the parameter budget

The point of Table 1 is not that BF-CONVUNEXT wins outright, because it does not, but where it sits on the accuracy/size trade-off. Figure 4 plots the four-set average PSNR at $\sigma = 25$ against approximate parameter count. BF-CONVUNEXT lands between the two size classes: comfortably above the sub-1M classic CNNs it beats, and within half a dB of models 8–10 $\times$ larger.

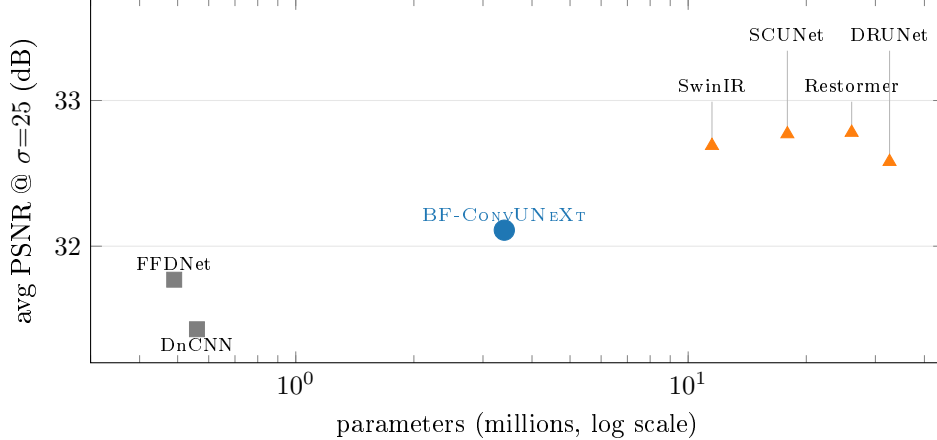


Figure 4: Four-set average PSNR at $\sigma=25$ vs. (approximate) parameter count. BF-CONVUNEXT (blue) sits on the efficient side of the trade-off: above the sub-1M CNNs and within ~ 0.5 – 0.7 dB of the 12–33M state of the art at $\sim 1/8$ – $1/10$ of their size. Parameter counts are approximate and collated from the respective papers.

The parameter count is only half the efficiency story; Table 3 adds compute and memory, measured on one RTX 4090. At 256^2 the model is ~ 104 GFLOPs and peaks at ~ 1.1 GB of GPU memory; cost scales linearly with pixel count (a fully-convolutional model), so a 512^2 image is $\sim 4\times$ the FLOPs and $\sim 3.7\times$ the peak memory. A single-image forward pass takes ~ 190 ms in eager `fp32` on this GPU (batched throughput is substantially higher). The model runs `fp32` with XLA by choice: we found mixed precision *slower* on this convolution-heavy U-Net, because the decoder’s bilinear-upsample gradient forces XLA off and the lost fusion outweighs the `fp16` tensor-core gain — so the “compact” claim is about parameters, FLOPs, and memory, and does not lean on a low-precision trick.

Table 3: Efficiency of BF-CONVUNEXT, measured on one RTX 4090 (single-image forward pass, `fp32`). FLOPs are per forward pass; memory is the peak GPU allocation. Cost is linear in pixel count.

Resolution	Parameters	FLOPs	FLOPs/megapixel	Peak GPU mem
256×256	3.39M	104 GFLOP	1.59 TFLOP	1.06 GB
512×512	3.39M	416 GFLOP	1.59 TFLOP	3.92 GB

5.6 Per-pixel uncertainty and coverage

Because the residual is a local score estimate, a natural question is whether a simple post-hoc interval calibrates. We build split-conformal per-pixel intervals (per- σ Mondrian, target coverage 0.90, $\alpha = 0.1$): calibrate a per- σ radius q_σ on a held-out calibration split and report empirical coverage on a disjoint test split. An early version of this study used only 32 calibration images and undercovered at ~ 0.877 (about 2.3 points below target), which we had wrongly attributed to an inter-image exchangeability gap. That attribution was a mistake, and it is worth correcting cleanly, because it turns on a subtlety about what “sample size” means for pixel-wise conformal.

The correct diagnosis is finite-sample size, but counted in *images*, not pixels. Pixels within one image are strongly correlated, so 32 images is a small effective sample even though it is ~ 6.3 M pixels. Re-measuring with 500 held-out COCO-validation calibration images and 500 test images settles it (Table 4): the plain per- σ scheme reaches 0.896–0.898, within ~ 0.3 points of nominal. To check whether the remaining gap is inter-image heteroscedasticity, we also ran a *variance-scaled* score — each pixel’s residual divided by a ground-truth-free per-image scale $\bar{u}_j = \text{mean}_i |D(y_j)_i - y_{j,i}|$, giving the interval $D(y)_i \pm q_\sigma \bar{u}_j$. It lands at the same ~ 0.897 with essentially identical (marginally wider) intervals, so there is no meaningful per-image scale

heterogeneity for it to exploit. The residual ~ 0.3 -point shortfall from 0.90 is consistent with mild within-image pixel correlation, not an image-level effect. In short: split-conformal calibration of this denoiser is straightforward given an adequate calibration set; the earlier undercoverage was an artifact of a 32-image pool, nothing deeper.

Table 4: Split-conformal per-pixel coverage (target 0.90, $\alpha=0.1$), re-measured at $n=500$ held-out COCO-validation calibration and 500 test images (one 256^2 crop per image). Two nonconformity scores: the per- σ absolute residual, and a ground-truth-free variance-scaled residual. The earlier 32-image pool undercovered at ~ 0.877 ; at $n=500$ both reach ~ 0.897 .

σ_{norm}	test PSNR (dB)	per- σ residual		variance-scaled	
		coverage	width	coverage	width
0.05	33.62	0.8962	0.068	0.8966	0.069
0.10	30.32	0.8970	0.098	0.8969	0.099
0.15	28.49	0.8973	0.119	0.8973	0.120
0.20	27.23	0.8976	0.136	0.8975	0.137
0.25	26.28	0.8980	0.151	0.8979	0.152

We stress the scope of this result: it says the denoiser admits well-calibrated *marginal* per-pixel intervals from a modest held-out set. It does *not* claim a calibrated Bayesian posterior or a global prior — that is a separate, genuinely negative result (the non-conservative Jacobian of Section 3), which conformal calibration does not touch.

6 Discussion: limitations and what does not work

Where it wins, and where it does not. As a compact, blind denoiser BF-CONVUNEXT is strong: it beats the classic sub-1M CNNs everywhere and closes most of the gap to models an order of magnitude larger, from a single checkpoint with no noise-level input. It does not beat the heavyweight state of the art, and the gap is structurally largest on Urban100, where non-local self-similarity rewards the attention and large receptive fields that a two-level local conv U-Net lacks.

Limits of the theory. The one genuinely negative result is the theory limit of Section 3: the learned residual is a local score, not the gradient of a global density, so uses that require a conservative denoiser — provable plug-and-play/RED convergence, energy-based sampling toward a fixed target, a calibrated Bayesian posterior — are out of reach. (This is orthogonal to the per-pixel conformal coverage of Section 5.6, which *is* well calibrated: frequentist marginal coverage of a fixed predictor needs no such property.) The homogeneity that does hold is inference-only and must be re-verified per checkpoint.

What the local score still buys. That the residual is only a local score, not a global prior, is less limiting than it sounds. In the annealed coarse-to-fine stochastic ascent of Kadkhodaie & Simoncelli [2021], the denoiser residual is used directly as the score at each effective noise level — $y \leftarrow y + h_t d_t + \gamma_t z_t$ with $d_t = (I - MM^\top) f(y) + M(m - M^\top y)$ for a linear measurement operator M with orthonormal columns — and this loop nowhere assumes a symmetric Jacobian or a global energy. The same frozen model, unchanged, therefore drives unconstrained sampling from the implicit prior (the $M=0$ case) and solves a family of linear inverse problems — inpainting, random-pixel recovery, super-resolution, spectral deblurring, compressive sensing — from one blind denoiser. Unlike plug-and-play/RED it offers no fixed-point convergence guarantee and needs no conservative operator; it is an empirical sampler. But it means the very property this section is careful *not* to oversell already has a substantial downstream reach.

Measurement limits. We report PSNR and SSIM for BF-CONVUNEXT (Tables 1 and 2), but the comparison against baselines is PSNR-only, because our collated source [Zhang et al., 2022] reports PSNR

and cross-source SSIM is protocol-inconsistent; a perceptual metric such as LPIPS is not measured at all. Table 1 shows point PSNR without per-cell error bars: with a fixed noise seed each cell is a deterministic function of the frozen model, so the relevant uncertainty is the across-image spread, which the DIV2K sweep reports as $\pm 0.4\text{--}0.8\text{ dB}$ 95% bootstrap CIs (Figure 2) — indicative of, but not measured on, the Table 1 leaderboard sets; per-cell CIs on Table 1 and a multi-seed estimate over noise realizations are left to future work. All results are from a single checkpoint, and we present no ablations: the individual contributions of the frozen Gabor stem, the Laplacian high-frequency skips, the `high_freq_blocks`, the bias-free normalization, and the noise curriculum are not isolated here. CBSD68 and McMaster are the two standard color sets we could evaluate on locally alongside Kodak24 and Urban100, and grayscale Set12/BSD68 would require a one-channel model. Each of these is deliberate scope, and each is the obvious next experiment.

Future work. The single most decisive follow-up is an **in-house bias-free baseline**: a bias-free flat DnCNN-style CNN (BF-CNN, in the sense of Mohan et al. [2020]) trained under the identical recipe. It would serve two roles at once — the closest-competitor row that our leaderboard comparison currently lacks (all baselines there are externally collated), and an *architecture* control isolating the value of the ConvNeXt-U-Net + Gabor + Laplacian integration over a plain bias-free convolutional stack. One caveat on framing: BF-CNN is itself bias-free, so it is *not* a test of bias-freedom; the clean test of the bias-free choice is our own network trained *with* biases (and with LayerNorm in place of the variance-only norm), which we also leave to future work. Beyond that: (i) a controlled ablation isolating each component’s dB contribution (Gabor stem, Laplacian skips, `high_freq_blocks`, bias-free norm, curriculum) with multi-seed variance; (ii) a perceptual metric (e.g. LPIPS) alongside the PSNR/SSIM already reported, and baseline SSIM under a single re-run protocol so the comparison can extend beyond PSNR; (iii) larger-capacity and deeper variants to probe how much of the SOTA gap is capacity vs. architecture; and (iv) whether a symmetrized or explicitly conservative variant recovers the plug-and-play guarantees the current model forgoes, and at what accuracy cost.

7 Related work

Bias-free denoising. Mohan et al. [2020] introduced bias-free CNNs for blind denoising and established the degree-1-homogeneity argument we build on. The residual-as-score reading rests on the classical Miyasawa/Tweedie identity [Miyasawa, 1961, Robbins, 1956], which also underpins the implicit-prior sampling line of Kadkhodaie & Simoncelli [2021]. BF-CONVUNEXT applies the bias-free discipline end-to-end to a modern block and front end.

CNN and transformer denoisers. DnCNN [Zhang et al., 2017] established the residual-CNN denoiser; FFDNet [Zhang et al., 2018] added a noise-level map for non-blind flexibility; DRUNet [Zhang et al., 2021] coupled a U-Net denoiser with plug-and-play restoration; SwinIR [Liang et al., 2021], Restormer [Zamir et al., 2022], and SCUNet [Zhang et al., 2022] brought (windowed/channel) attention and are the current leaderboard leaders. These are our baselines; all except our model and DnCNN’s blind variant either use a noise map or are larger by $8\text{--}60\times$.

Building blocks. The block is a CONVNEXT [Liu et al., 2022] in a U-Net [Ronneberger et al., 2015]. The multi-scale skip decomposition is a Laplacian pyramid [Burt & Adelson, 1983], and the frozen front end is a Gabor filter bank [Gabor, 1946]. BF-CONVUNEXT contributes their bias-preserving integration and its measurement, not the pieces.

8 Reproducibility

The evaluated checkpoint is `convunext_denoiser_base` (3,393,936 parameters; depth 2; widths 66/132/264; three CONVNEXT-V1 blocks per scope plus three per high-frequency skip band; frozen 32-filter, 7×7 Gabor stem; BIASFREEBATCHNORM; LeakyReLU(0.1); linear grouped head). It was trained with the recipe of Section 4. The full-image benchmark uses whole test images reflect-padded to a multiple of 16, PSNR at `max_val` = 1.0 on $[-0.5, 0.5]$, on the standard CBSD68 / Kodak24 / McMaster / Urban100 sets. All

numbers in Table 1, Table 2 and Figure 2 are measured from that single checkpoint; baseline numbers are the published values as collated by Zhang et al. [2022]. The coverage study (Table 4) uses 500 held-out COCO-val2017 calibration and 500 test crops (256², one per image); the efficiency numbers (Table 3) are single-image fp32 forward passes on one RTX 4090.

Gabor stem, in full. The frozen stem is fully specified so it can be regenerated deterministically (given the grid and seed below; the exact per-axis `linspace` resolution and channel-ordering are fixed in the released code). It is a depthwise 7×7 convolution with depth multiplier 32 (so 3×32=96 output channels), `use_bias=False`, `trainable=False`. Each output channel is one real 2D Gabor wavelet $g(u, v) = \exp\left(-\frac{u_\theta^2 + \gamma^2 v_\theta^2}{2s^2}\right) \cos\left(\frac{2\pi u_\theta}{\lambda} + \psi\right)$, with (u_θ, v_θ) the coordinates rotated by θ . The five parameters are swept on a regular `linspace` grid over the Özbek–Ekenel Table-I ranges — $s \in [2, 21]$, $\theta \in [0^\circ, 360^\circ]$, $\lambda \in [8, 100]$, $\gamma \in [0, 300\%]$, $\psi \in [0^\circ, 360^\circ]$ — one distinct $(s, \theta, \lambda, \gamma, \psi)$ tuple per output channel, replicated identically across the three input channels, with global seed 42. Because the bank is fixed, it contributes zero trainable parameters and requires no checkpointing beyond these constants.

9 Conclusion

We described BF-CONVUNEXT, a compact bias-free CONVNEXT U-Net that integrates a frozen Gabor stem, Laplacian-pyramid high-frequency skips, and end-to-end degree-1 homogeneity, and we measured it against the color-denoising leaderboard. As a single 3.4M-parameter blind model it beats the classic sub-1M CNNs on all four standard sets at all three noise levels and comes within about half a dB of state-of-the-art models an order of magnitude larger, with the residual gap concentrated on self-similar imagery. Within its stated limits — a single, non-conservative checkpoint — it is a useful data point on how much denoising accuracy, and, through the implicit-prior score, how much downstream inverse-problem capability, is reachable from a small, strictly bias-free model.

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